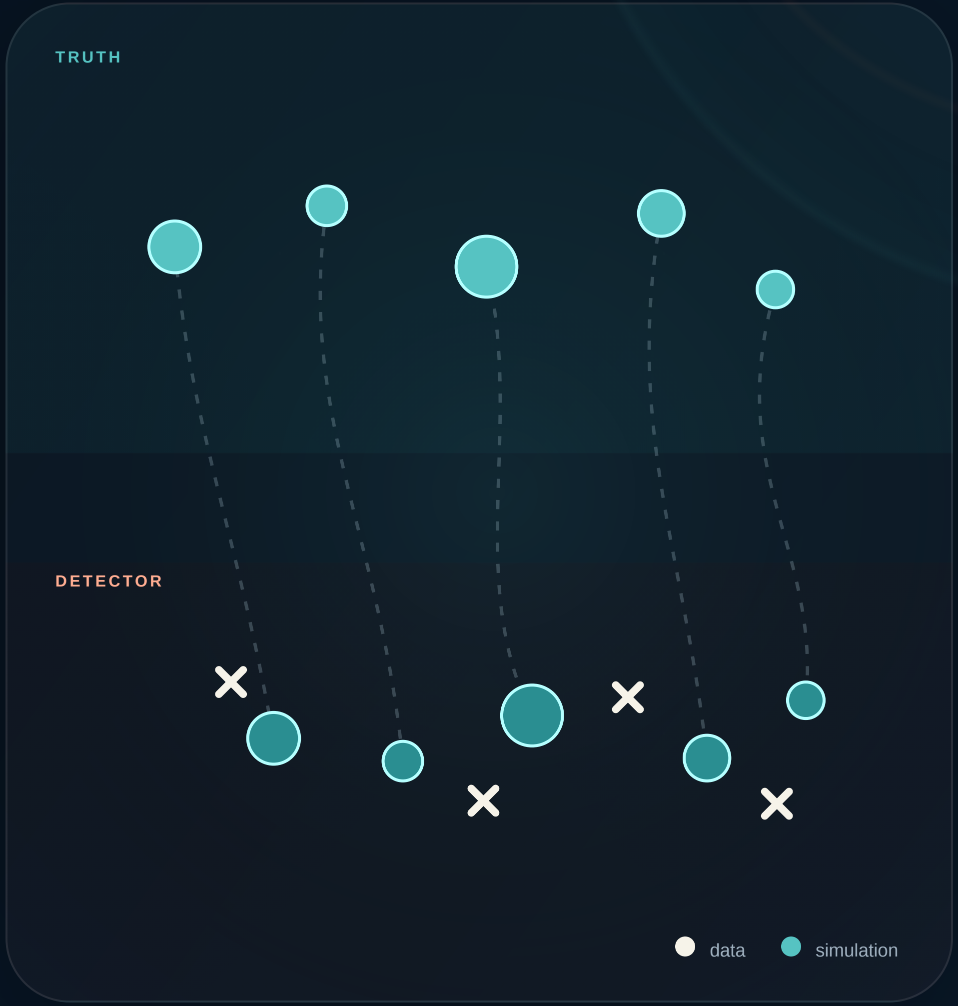
MINERVA COLLABORATION · 16 JULY 2026

OmniFold with MINERvA open data

Learning weights on events instead of corrections between bins

Joseph Bailey · MINERvA ME FHC CC-inclusive

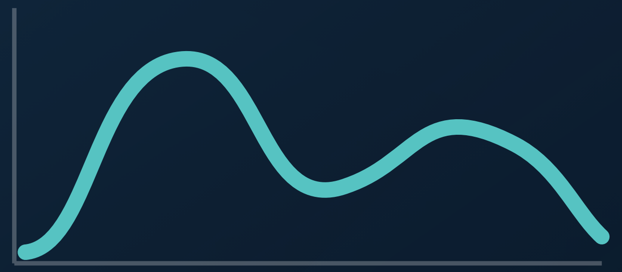


THE MEASUREMENT PROBLEM

The detector shows a distorted image of nature

WHAT WE WANT

truth-level cross section



detector

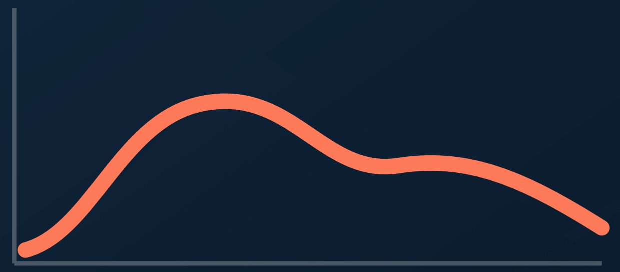
smearing

inefficiency

background

WHAT WE OBSERVE

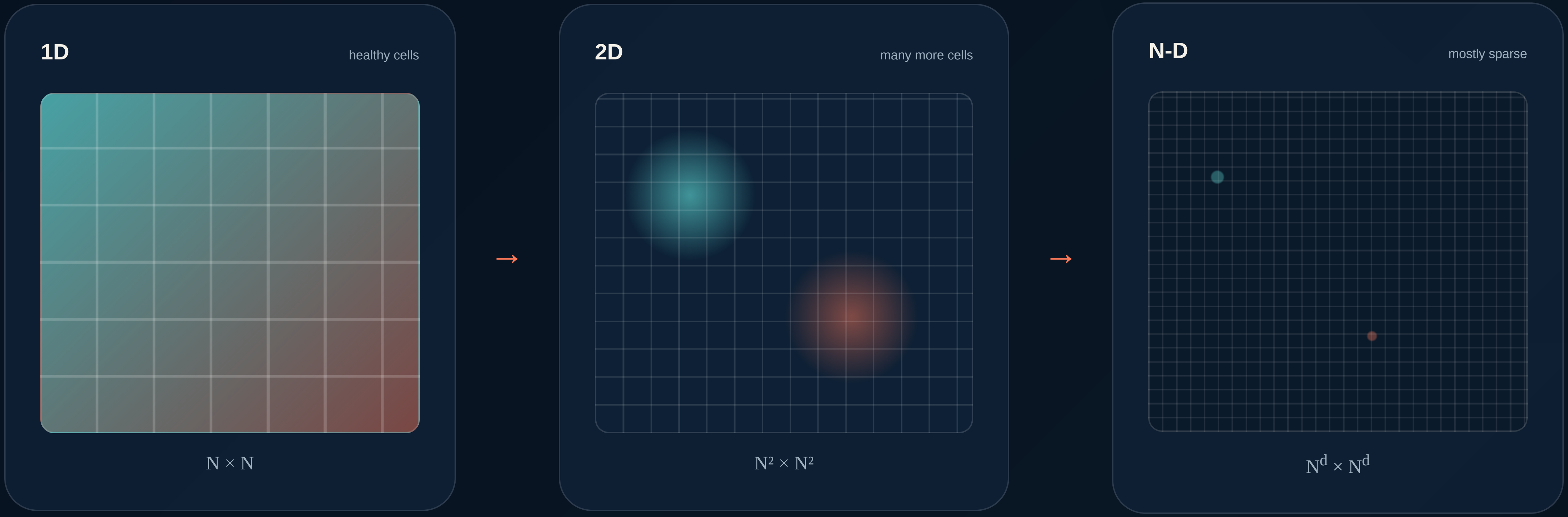
selected reconstructed events



Unfolding asks: which truth distribution could have produced the data we selected?

WHY UNBINNED?

Adding observables multiplies response cells



The events were never sparse.

The histogramming made them sparse.

THE ENGINE

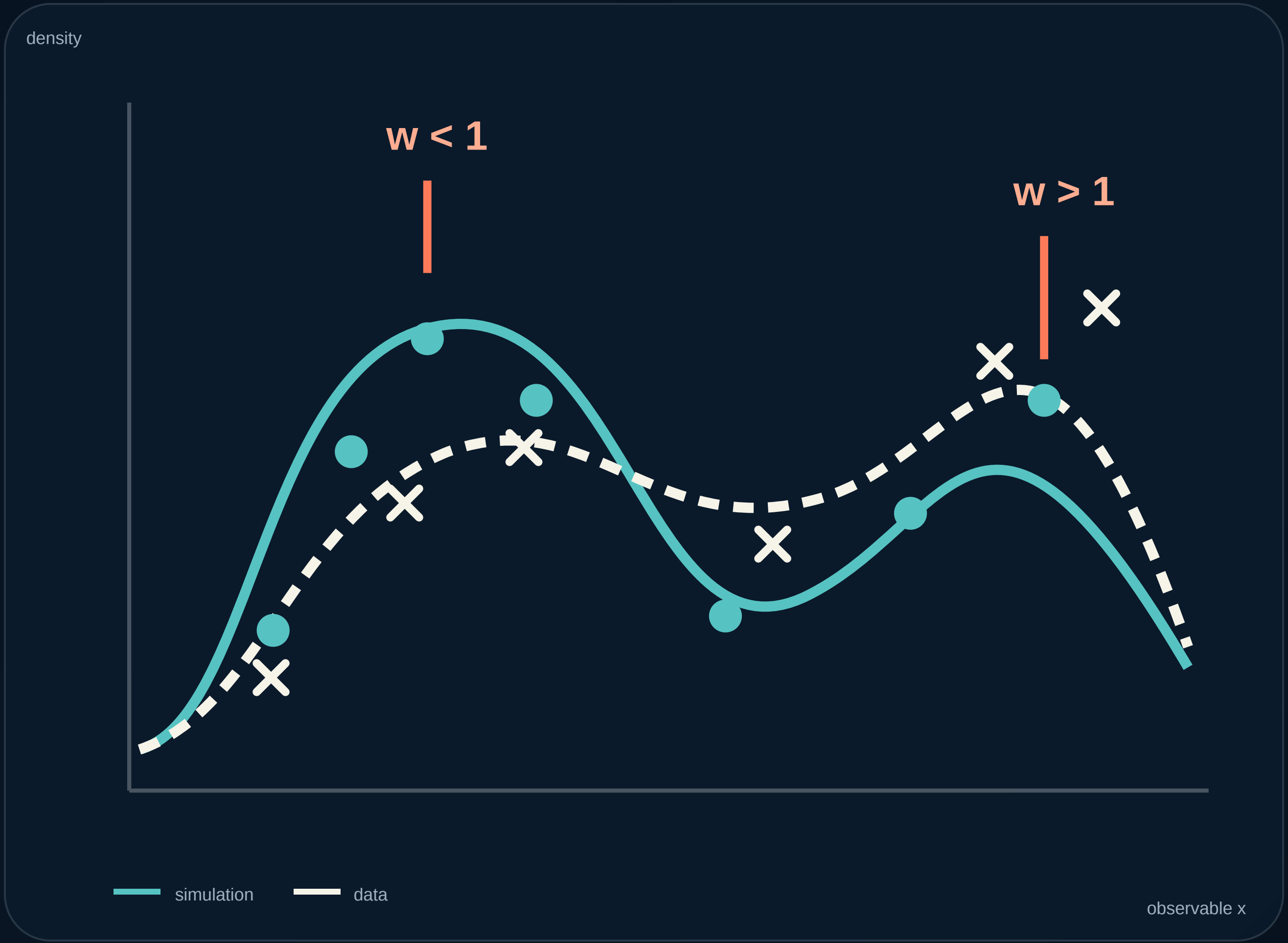
A classifier is a ratio estimator

Ask a classifier to distinguish data from simulation at a point x.

WITH CLASS-NORMALIZED TRAINING

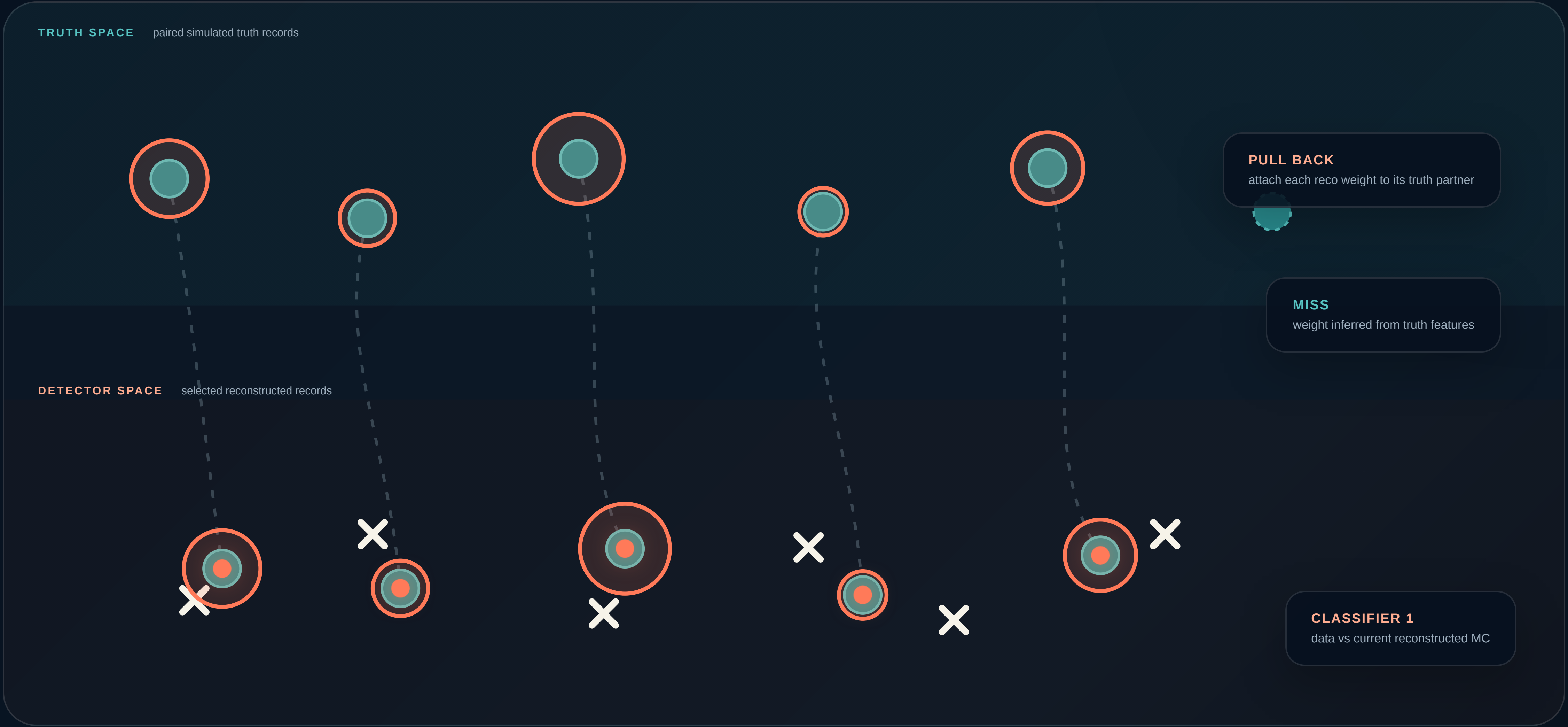
$$w(x) = \frac{p(\text{data} \mid x)}{1 - p(\text{data} \mid x)}$$

Hard to tell apart → weight near 1
Data-like region → weight above 1
MC-heavy region → weight below 1



ONE OMNIFOLD ITERATION · STEP 1

Match the data at detector level



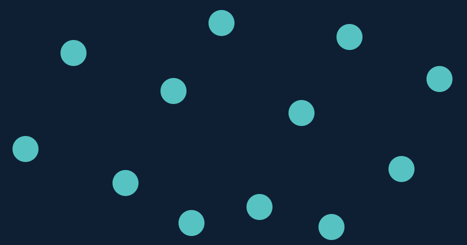
ONE OMNIFOLD ITERATION · STEP 2

Turn the pullback into a truth-level function

A

Original truth MC

prior event weights



vs

B

Pulled-back truth MC

Step-1 event weights

CLASSIFIER 2

smooth truth-space ratio

NEXT ITERATION

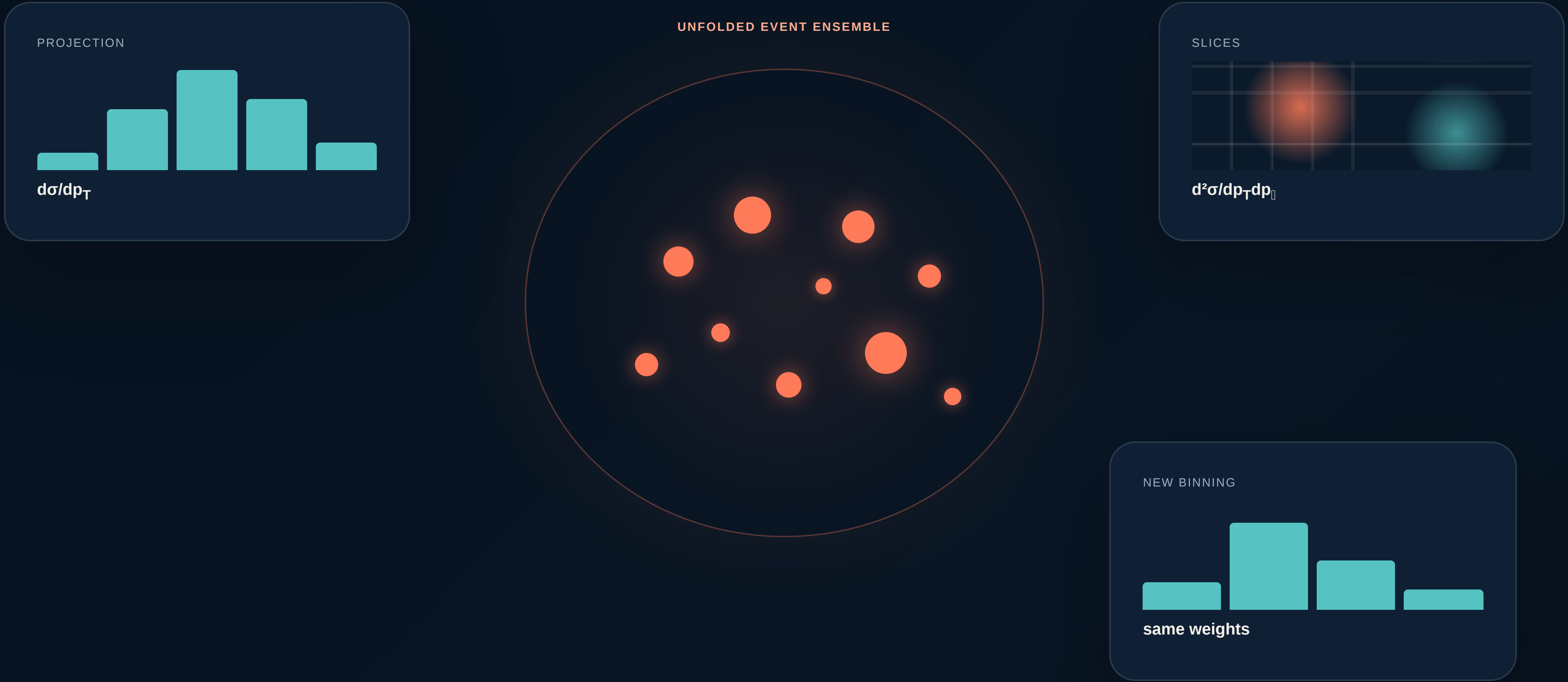
new push weights

push to detector ▯ repeat

Two classifiers per iteration: **detector correction** + **truth regularization**

THE OUTPUT

One weighted truth ensemble, many views



Rebin or project recorded truth features without retraining. Adding a genuinely new feature requires it in the event representation.

FROM MINERVA OPEN DATA TO OMNIFOLD

Four event trees preserve the analysis logic

DATA

data

selected reconstructed events

4,119,797

MC

mc_background

wrong-sign, NC and out-of-fiducial background

MC · PAIRED

mc_signal_reco

reco and truth features together

MC · AUTHORITATIVE

mc_truth_denom

every true signal event

32,849,103

↓ ↓ ↓ ↓

STEP 1 TARGET

data − genuine background − signal fakes ↔ reconstructed signal MC

PAIRING + MISSES

truth-authoritative keys; truth-only events stay in the sample

ME1A 1B 1C 1D 1E 1F 1G 1L 1M 1N 1O 1P

run separately → merge

FROM EVENT WEIGHTS TO A CROSS SECTION

The learned weight is only the middle of the measurement

01

phase-space data

$0 \leq p_T \leq 4.5$

$1.5 \leq p_{\square} \leq 60$ GeV/c

→

02

purity weight

non-negative

(data − background)/data

→

03

OmniFold

learn truth-level

push weights

→

04

truth yield

$\Sigma\, w_{push}\, w_{truth}$

inside each final bin

$$d^2\sigma_b = \frac{\Sigma_{i \square b}\, w_{push,i} w_{truth,i}}{c_b \cdot \Phi \cdot N_{POT} \cdot N_{nucleons} \cdot \Delta p_{T,b} \cdot \Delta p_{\square,b}}$$

flux, POT and target normalization

completeness = 1 by construction

× no second absolute-efficiency division

TRY TO BREAK IT BEFORE TRUSTING IT

Validation probes different failure modes

Closure

truth reweights, hidden resolution variable, alternate model

PASS

2D pull diagnostic

|r|

same-ensemble Gaussianity check

0.794

unit-normal reference 0.798; not coverage

Iterations

5→10

doubling the iteration count

0.026%

change in total σ

Classifier family

GBDT

neural-network cross-check

1.0078

NN / GBDT total

Binned limit

IBU

first update recovers D'Agostini behavior

CONNECTED

Bottom line

AUC

descriptive classifier check

0.535 → 0.501

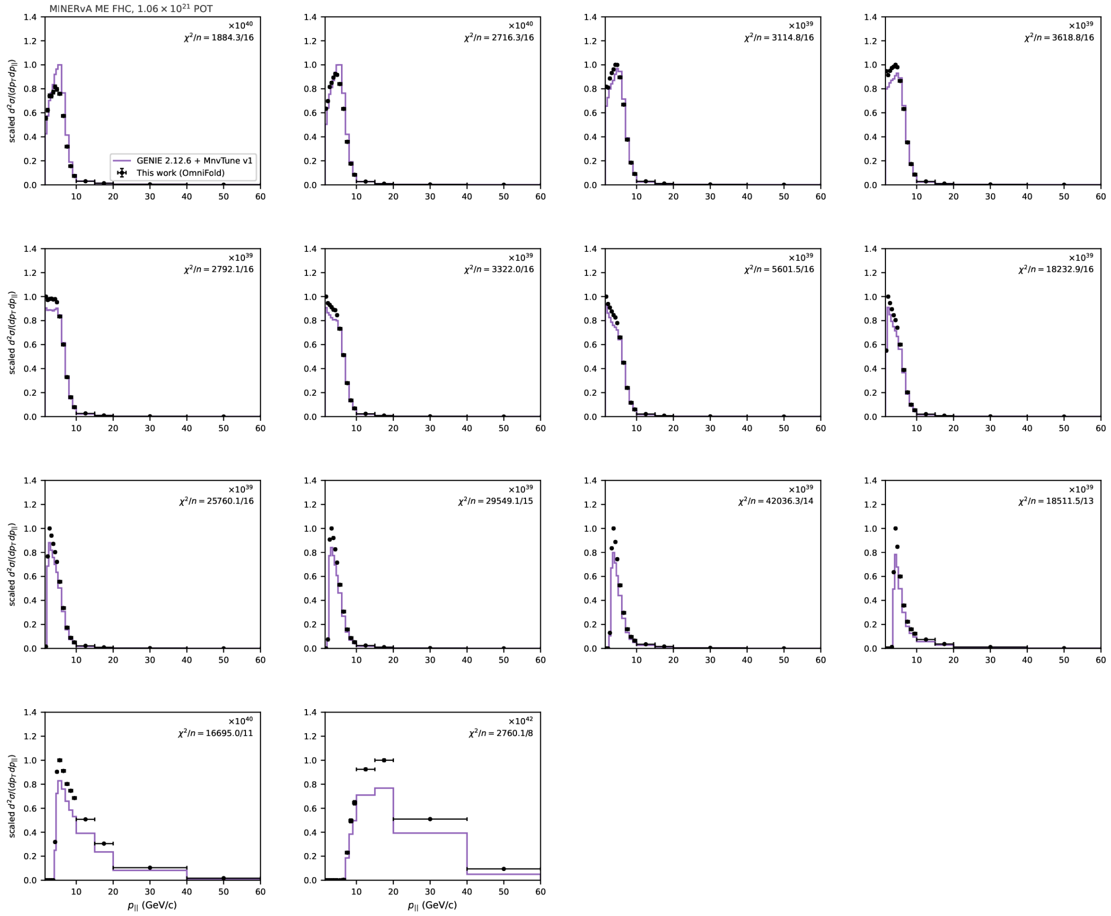
no calibrated p-value

THE TRUST ANCHOR

Same data.
Nearly the same answer.

Unbinned in training; binned afterward into MINERvA's published 14 × 16 grid.

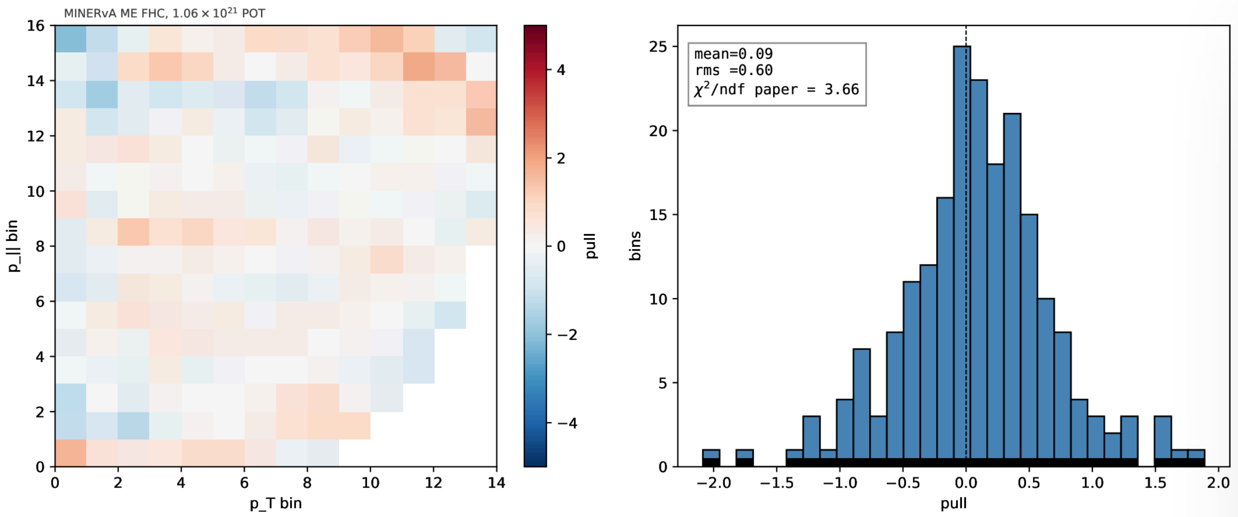
1.011	total σ ratio OmniFold / paper
1.006	median bin ratio
94.1%	of reported bins within 10%



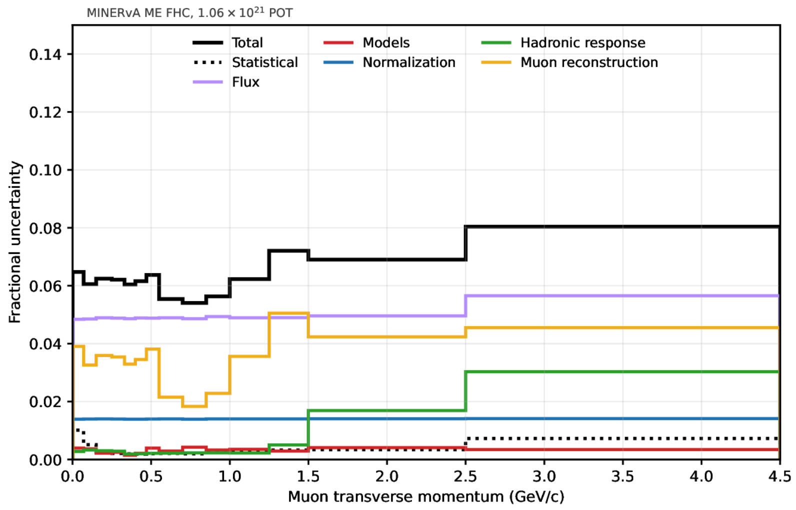
ME FHC CC-inclusive · 205 reported bins

AGREEMENT IS GOOD — BUT NOT FEATURELESS

Per-bin agreement and correlated shape tell different stories



Published-σ RMS 0.60 descriptive per-bin scale; shared data

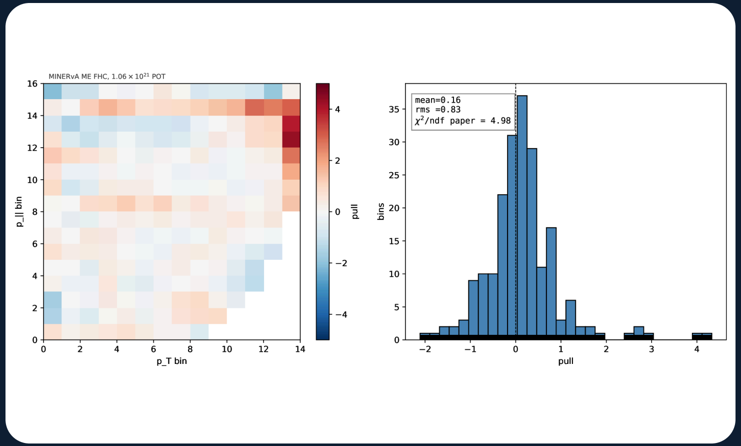


6.87% vs 6.86% median uncertainty: this work vs paper

Indicative paper-covariance distance: 3.66/bin correlated low-p_T shape modes; no shared-data cross-covariance

WHY KEEP THE EVENTS?

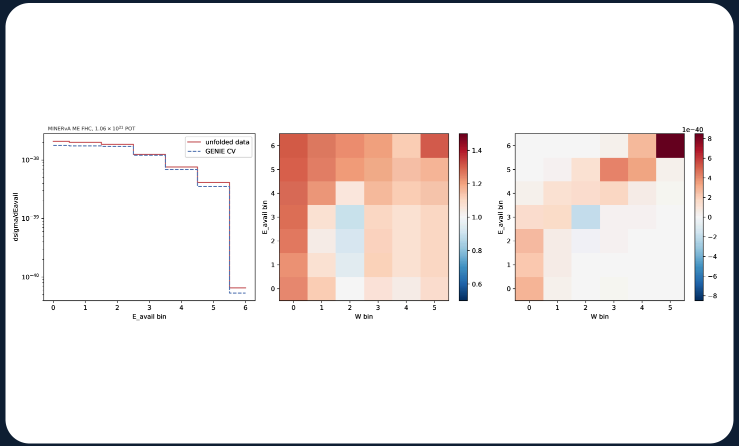
A reusable measurement object opens new questions



VALIDATION

Add E_{avail}, q_3, W

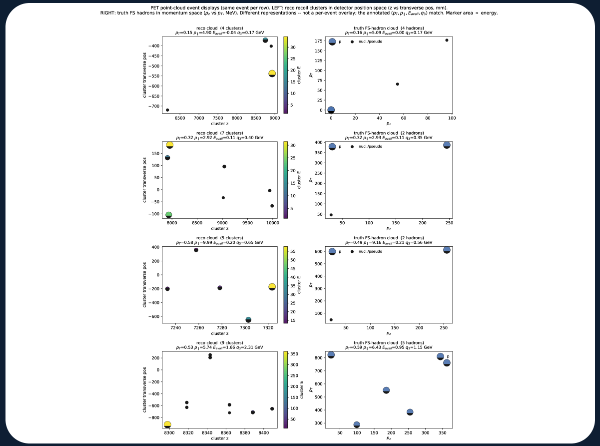
Every new dimension must recover its lower-dimensional anchor when marginalized.



CENTRAL-VALUE OBSERVATION

Localize discrepancies

The positive difference concentrates at high E_{avail} and high W. Significance is withheld pending corrected UQ.



OUTLOOK

Learn from event clouds

Replace a few scalar inputs with reconstructed calorimeter-cluster point clouds.

TAKEAWAYS

Learn weights on events.
Bin only when needed.

01 OmniFold is iterative density-ratio reweighting through paired simulated events.

02 The MINERvA open product supports a complete event-level cross-section pipeline.

03 The 2D result reproduces the published measurement and anchors higher-dimensional work.

DISCUSSION

Which projections—and which validation tests—would be most useful to MINERvA?